昊邦电子
HAOBANG-PCB.COM

QR-ME-15(A)

Engineering Confirmation Note
Layer stackup & impedance

Subject :Questions about the stack and the impedance value

From : ShenZhen HaoBang Circuit Co., Ltd
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Cus NO : STK-94921
DATE : 2024/10/31 12:48:02

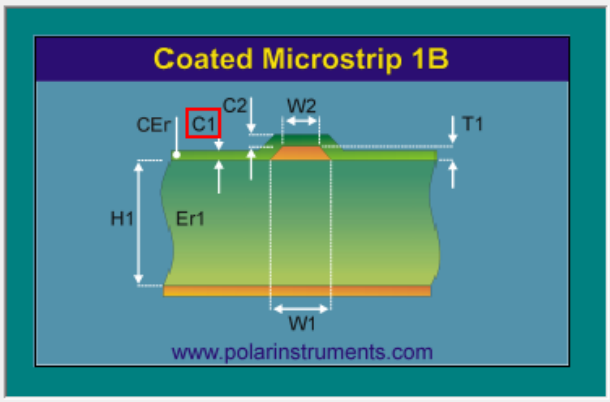
Stackup Information:

Layer	Dk	Info	Df	Thickness
SMT	**3.4**	==========	0.02	0.5 mil
TOP		==========		1/2 Oz +Plating=1.7mil
	4.0	PP	0.018	3.8 (mil)
L2		==========		1 Oz=1.2mil
	4.2	Core	0.018	4.0(mil)
L3		==========		1 Oz=1.2mil
	4.2	Core+pp	0.018	37.4(mil)
L4		==========		1 Oz=1.2mil
	4.2	Core	0.018	4.0(mil)
L5		==========		1 Oz=1.2mil
	4.0	PP	0.018	3.8 (mil)
BOT		==========		1/2 Oz+Plating=1.7mil
SMB	**3.4**	==========	0.02	0.5 mil
Finished:		62.2(+6.3/-6.3) mil		1.59(+0.16/-0.16) MM
Designed:		63mil		1.6 MM
Material:		TU-768		TU-768

Impedance Information:

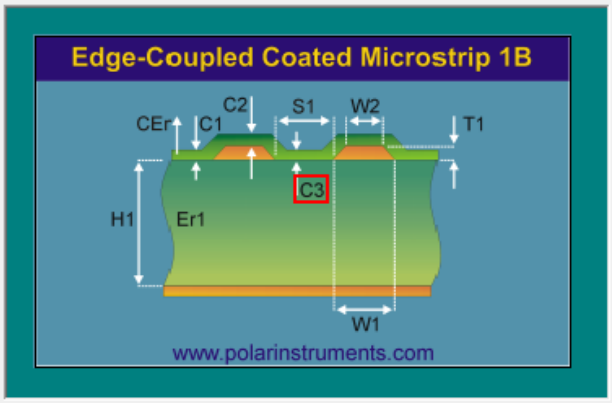
Ctrl	Ref	Imp_type	Cust_req	Imp_req	HB_des	Imp_des	mask	H1	Er1	H2	Er2
L1	L2	Single-Ended	6.0	50+/-10%	6.0	50.03	Yes	3.8	4.0		
L1	L2	Difference	6.0/7.5	90+/-10%	6.0/7.5	89.65	Yes	3.8	4.0		
L1	L2	Difference	4.7/8.0	100+/-10%	4.7/8.0	100.46	Yes	3.8	4.0		
L3	L2/L5	Single-Ended	5.0	50+/-10%	5.0	49.94		4.0	4.2	43.8	4.2
L4	L2/L5	Single-Ended	5.0	50+/-10%	5.0	49.94		4.0	4.2	43.8	4.2
L6	L5	Single-Ended	6.0	50+/-10%	6.0	50.03	Yes	3.8	4.0		
L6	L5	Difference	6.0/7.5	90+/-10%	6.0/7.5	89.65	Yes	3.8	4.0		
L6	L5	Difference	4.7/8.0	100+/-10%	4.7/8.0	100.46	Yes	3.8	4.0		

Coated Microstrip 1B



				Tolerance	Minimum	Maximum	
Substrate 1 Height	H1	3.8000	+/-	0.0000	3.8000	3.8000	Calculate
Substrate 1 Dielectric	Er1	4.0000	+/-	0.0000	4.0000	4.0000	Calculate
Lower Trace Width	W1	6.0000	+/-	0.0000	6.0000	6.0000	
Upper Trace Width	W2	5.2000	+/-	0.0000	5.2000	5.2000	Calculate
Trace Thickness	T1	1.7000	+/-	0.0000	1.7000	1.7000	Calculate
Coating Above Substrate	C1	1.7000	+/-	0.0000	1.7000	1.7000	
Coating Above Trace	C2	0.5000	+/-	0.0000	0.5000	0.5000	
Coating Dielectric	CEr	3.4000	+/-	0.0000	3.4000	3.4000	
Impedance	Zo	50.03			50.03	50.03	Calculate

Edge-Coupled Coated Microstrip 1B



				Tolerance	Minimum	Maximum	
Substrate 1 Height	H1	3.8000	+/-	0.0000	3.8000	3.8000	Calculate
Substrate 1 Dielectric	Er1	4.0000	+/-	0.0000	4.0000	4.0000	Calculate
Lower Trace Width	W1	6.0000	+/-	0.0000	6.0000	6.0000	
Upper Trace Width	W2	5.2000	+/-	0.0000	5.2000	5.2000	Calculate
Trace Separation	S1	7.5000	+/-	0.0000	7.5000	7.5000	Calculate
Trace Thickness	T1	1.7000	+/-	0.0000	1.7000	1.7000	Calculate
Coating Above Substrate	C1	1.7000	+/-	0.0000	1.7000	1.7000	
Coating Above Trace	C2	0.5000	+/-	0.0000	0.5000	0.5000	
Coating Between Traces	C3	1.7000	+/-	0.0000	1.7000	1.7000	
Coating Dielectric	CEr	3.4000	+/-	0.0000	3.4000	3.4000	
Differential Impedance	Zdiff	89.65			89.65	89.65	Calculate

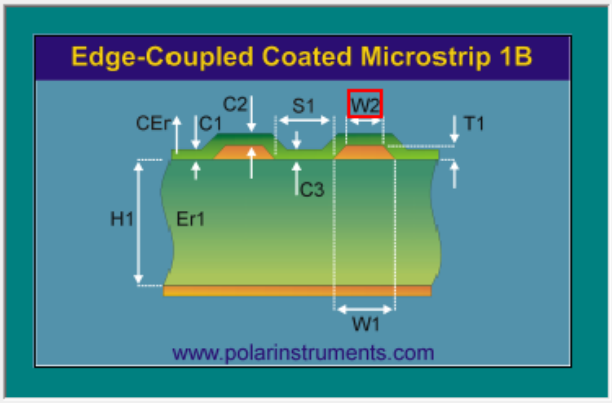
Notes

Add your comments here

Interface Style

Standard

Edge-Coupled Coated Microstrip 1B



				Tolerance	Minimum	Maximum	
Substrate 1 Height	H1	3.8000	+/-	0.0000	3.8000	3.8000	Calculate
Substrate 1 Dielectric	Er1	4.0000	+/-	0.0000	4.0000	4.0000	Calculate
Lower Trace Width	W1	4.7000	+/-	0.0000	4.7000	4.7000	
Upper Trace Width	W2	3.9000	+/-	0.0000	3.9000	3.9000	Calculate
Trace Separation	S1	8.0000	+/-	0.0000	8.0000	8.0000	Calculate
Trace Thickness	T1	1.7000	+/-	0.0000	1.7000	1.7000	Calculate
Coating Above Substrate	C1	1.7000	+/-	0.0000	1.7000	1.7000	
Coating Above Trace	C2	0.5000	+/-	0.0000	0.5000	0.5000	
Coating Between Traces	C3	1.7000	+/-	0.0000	1.7000	1.7000	
Coating Dielectric	CEr	3.4000	+/-	0.0000	3.4000	3.4000	
Differential Impedance	Zdiff	100.45			100.45	100.45	Calculate

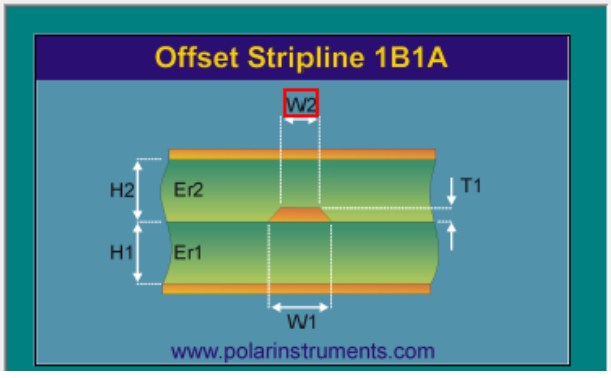
Notes

Add your comments here

Interface Style

Standard

Offset Stripline 1B1A



				Tolerance	Minimum	Maximum	
Substrate 1 Height	H1	4.0000	+/-	0.0000	4.0000	4.0000	Calculate
Substrate 1 Dielectric	Er1	4.2000	+/-	0.0000	4.2000	4.2000	Calculate
Substrate 2 Height	H2	43.8000	+/-	0.0000	43.8000	43.8000	Calculate
Substrate 2 Dielectric	Er2	4.2000	+/-	0.0000	4.2000	4.2000	Calculate
Lower Trace Width	W1	5.0000	+/-	0.0000	5.0000	5.0000	
Upper Trace Width	W2	4.2000	+/-	0.0000	4.2000	4.2000	Calculate
Trace Thickness	T1	1.2000	+/-	0.0000	1.2000	1.2000	Calculate
Impedance	Zo	49.94			49.94	49.94	Calculate

Remark ：

Customer Reply：　　Accept:　　Disapprove:

Date:：　　Signature：

Mark ：

Remark ：